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24 March 1967
SUPERSEDING
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MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED, M6, ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated, M6, Assembly loaded with Cartridge, Thruster, M67A1 for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION
Military
- MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS
Military
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions
- MIL-STD-1167 - Ammunition Data Card
- MS28741 - Hose Assembly, Detachable End-Fitting, Medium Pressure

DRAWINGS
Munitions Command
- F8594149 - Thruster, Cartridge Actuated, M6, Assembly
- C8797437 - Box, Packing, Ammunition, for Thruster, Cartridge Actuated, M6, w/Ctg. M67, in Fiber Container, M363.
- C8797438 - Container, Ammunition, Fiber M363 for Thruster, Cartridge Actuated, M6, w/Ctg M67
- IEL-8594149 - Inspection Equipment List for Thruster, Cartridge Actuated, M6, Assembly

FSC 1377

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(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Torque.- Torque requirements shall be as specified in Table I. No movement or deformation shall occur upon application of the breakaway torque.

Table I

Component	A Seating or Bottoming (inch pounds)	B Breakaway (inch pounds) (minimum)
Body and Breech	100 + 20	400

3.3 Residual magnetism.- The thruster shall not deflect the indicator of the compass more than three degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional.

3.4.1 The thruster shall not fail to fire.

3.4.2 The thruster shall function without mechanical failure (see 6.1).

3.4.3 No load.- The thruster fired with no load on the piston shall complete its stroke of 1 7/16 inches minimum and shall not separate from the body of the thruster (see 6.2).

3.4.4 The thruster fired under load condition shall perform as follows (see 6.3).

3.4.4.1 Load.- The thruster shall propel a 60 pound weight horizontally through a distance of 1 7/16 inches minimum.

3.4.4.2 Thrust.- The initial thrust developed shall be 400 pounds minimum (see 6.3).

3.4.5 Temperature.- The thruster shall comply with all the ballistic requirements throughout the temperature range of minus 65 degrees to plus 165 degrees F.

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3.4.6 Function delay.- The time elapsed between the energizing of the thruster and the initial movement of the propelled weight, shall not exceed 50 milliseconds.

3.4.7 Initiation.- The thruster shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 20 feet of aircraft hose, size no. 4, MS28741. at 70° ± 5°F.

3.5 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- All components shall be free from cracks, splits, perforations or any similar defect.

3.5.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified therein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

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4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3
1801 - 10,000	200	2	3	3	4

4.2.2.1 Cartridge, Thruster, M67A1, Dwg. D8593295 covering a detail of Dwg. F8594149.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridge	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Foreign matter	Visual
MINOR	
201. Case scratched or dented	Visual
202. Varnish missing or inadequate	Visual

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4.2.2.2 Breech Sub-Assembly, consisting of components 8593754, 8593756, 8594506, 8594744, 501217 and set screw, covering a detail of Dwg. F8594149.

	Defect	Method of Inspection
CRITICAL		
1.	Striker hole obstructed	Visual
MAJOR		
101.	Thread damaged	Visual
102.	Protective finish inadequate or defective	Visual
103.	Burr	Visual
104.	Foreign matter	Visual
MINOR		
201.	Escapement hole missing	Visual
202.	Set screw above flush	Visual

4.2.2.3 Assembly, Dwg. F8594149 less Shipping Plug A8593299.

CRITICAL		
1.	Shear pin not engaged in firing pin	Gage
2.	Foreign matter in gas port	Visual
MAJOR		
101.	Total length	Gage
102.	Set screw above flush	Visual
103.	Set screws not filled with barrier material	Visual
104.	Thread damaged	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
201.	Protective finish inadequate or defective	Visual

4.2.2.4 Assembly, Dwg. F8594149.

CRITICAL		
1.	Improper assembly	X-ray

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4.2.2.4 (Cont'd)

Defect	Method of Inspection
CRITICAL	
2. Any part missing as detected by X-ray	X-ray
3. Metal defective	Visual
MAJOR	
101. Foreign matter	Visual
102. Marking missing, incorrect or illegible	Visual
MINOR	
201. Length to breech	Gage
202. Total length	Gage
203. Location of labels, incorrect	Visual
204. Shipping plug missing	Visual
205. Varnish missing or inadequate	Visual

4.2.2.5 Packing in Fiber Container, Dwg. F8797438.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.6 Packing in Wooden Box, Dwg. B8797437.

CRITICAL	
None defined	

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4.2.2.6 (Cont'd)

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual & Manual
203. Strapping missing, loose, broken or improperly assembled	Visual & Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.- Each body and breech assembly shall be subjected to a breakaway torque. Failure of any assembly to comply with the requirements of 3.2 and Table I shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism.- Each thruster shall be subjected to the test for excessive residual magnetism. Failure of any thruster to comply with the requirements of 3.3 shall be cause for rejection of the thruster.

4.2.3.3 Functional tests.- The following tests will be performed by the Government.

4.2.3.3.1 Ballistics.- Thirty two units shall be tested to determine compliance with the requirements of 3.4.1 through 3.4.7. The units shall be temperature conditioned as provided in Table III and 4.3.3.1.1 and 4.3.3.1.2. Failure of any unit to comply with the requirements shall be cause for rejection of the lot.

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Table III

Sample Size	Method	Temperature
10	Load (See 3.4.4.1)	-65° to -70°F
10	Load (See 3.4.4.1)	65° to 75°F
10	Load (See 3.4.4.1)	160° to 165°F
1	No Load (See 3.4.3)	-65° to -70°F
1	No Load (See 3.4.3)	160° to 165°F

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment in accordance with IEL-8594149.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.- A torque wrench, capable of measuring the torque requirements as specified in Table I shall be used with an appropriate adapter to determine the torque. The body shall be secured in a holding device and the torque applied to the breech.

4.3.2 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from and at the same height as the compass, in a north-south horizontal position with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be followed for each 90 degree rotation of the thruster.

4.3.3 Functional tests.

4.3.3.1 Ballistics.- The thrusters shall be temprature conditioned in accordance with Table III and the following:

4.3.3.1.1 Temperature conditioning.- A simulated thruster and cartridge with a temperature sensitive element, placed in the dummy cartridge shall be stored in a temperature conditioning chamber at the same time as the thrusters to be temperature conditioned. The temperature conditioning time shall be a minimum of one hour longer than the time necessary for the temperature of the simulated cartridge to reach the required temperature.

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4.3.3.1.2 Testing time and reconditioning.- Upon completion of the required temperature conditioning, the thruster shall be ballistically tested within 10 minutes after removal from the conditioning chamber. Thrusters shall be reconditioned if this time is exceeded. When reconditioning thrusters at minus 65 degrees F, all condensation shall be removed from the interior surfaces before returning the thruster to the conditioning chamber.

4.3.4 Method of testing.- The thruster tested under the conditions as specified herein, shall be mounted so that the weight will be propelled horizontally when fired, and supported rigidly so that the entire thrust will be borne by the thruster mounting points.

4.3.4.1 Thrust.- Secure the 60 pound weight to a stationary fixture with two 200 pound shear pins mounted for single shear each, or one 200 pound shear pin mounted for double shear. The thruster is mounted in the fixture in contact with the 60 pound weight. The shearing of the fixture pin, or pins, and completion of the piston stroke is evidence that the thrust developed by the propellant system complies with the functional requirements. A shock absorbing device should be incorporated into the test fixture to absorb impact energy of the moving weight. The thrust may be determined by any method approved by the contracting officer.

4.3.4.2 Delay time.- The function delay time shall be recorded in the following or equal manner: The start of the function delay time is to be recorded by energizing a piezo-electric gage in conjunction with a fixture. The shearing of the thruster shear pin in the breech shall be considered the start of the function delay time and is evidenced by a break in the continuous pressure rise on the pressure-time record of the entrance gas port pressure. The end of the function delay time is to be recorded by the initial actuation of a stroke-time device.

4.3.4.3 Stroke time.- The time to complete stroke shall be recorded in the following or equal manner: The thruster piston shall make contact with a spaced wire wound spool or actuate a variable resistance type device to give a stroke time relationship. The thrust shall be recorded by thrust or load cell apparatus.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with Drawings C8797437 and C8797438.

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5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 No load.- Permanent yielding of the trunnion mounting lugs and piston shall not be cause for rejection when thrusters are tested under no load condition.

6.3 Load.- The expected performance of the thruster under conditions simulating those existing in aircraft, shall be referred to as load condition in this specification.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

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